Biased H-LIP Control for Walking on Sloped and Accelerating Ground

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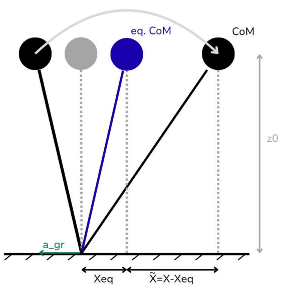
OVERVIEW

Motivated by humanoid walking on sloped terrain and accelerating ground, this work studies how effective gravito-inertial effects shift the natural balance behavior of legged robots. We propose a Biased H-LIP model that embeds slope-induced gravity projection and acceleration-induced inertial effects as a shifted equilibrium in the reduced-order dynamics. This preserves the standard H-LIP structure in shifted coordinates, enabling unified step-planning control with minimal structural modification.



Biased H-LIP interpretation. Slope or platform acceleration shifts the natural CoM equilibrium to x_{eq}. The current CoM is then described relative to this shifted point using $\tilde{x} = x - x_{eq}$.

Modeling Motivation

Underactuated humanoid walking is commonly modeled using a reduced-order Linear Inverted Pendulum formulation [1].This assumes CoM height is approximately constant at z_0, and the stance foot is a fixed contact during single support.

This model defines sagittal CoM state by its position x relative to the stance foot and velocity v, the nominal step length as u^*, and a step contains single step phase (SSP) T_s and double step phase (DSP) T_d.

On inertial ground, CoM's motion is naturally centered around the stance foot. However, on a non-inertial platform, an effective gravito-inertial acceleration shifts this natural center to x_{eq}, as shown in the reference diagram.

Our motivation is to regulate walking using the shifted equilibrium introduced by the gravito-inertial effect, so the environmental effect is embedded directly into the dynamics rather than treated only as an external disturbance. Therefore, we use an H-LIP-based step-planning model, to account for both the continuous CoM motion and actual discrete step updates.

Unified Biased H-LIP Model

Effective acceleration:

$$g_{eff} = \begin{bmatrix} g_{eff,x} \\ g_{eff,z} \end{bmatrix}$$

, where $g_{eff,x}$ is the sagittal bias and $g_{eff,z}$ is the normal component.

Biased CoM dynamics: $\ddot{x} = \frac{g_{eff,z}}{z0} x - g_{eff,x}$

Shifted equilibrium: $x_{eq} = z_0 \frac{g_{eff,x}}{g_{eff,z}}$

Recovered H-LIP form: : $\ddot{\tilde{x}} = \alpha^2 \tilde{x}$

, where $\alpha = \sqrt{\frac{g_{eff,z}}{z_0}}$

Why this structure is applicable:

Preserve the H-LIP phase-space orbit: $E(x, v) = v^2 - (\alpha\tilde{x})^2$; which remains constant: $\frac{\partial E}{\partial t} = 0$

The shifted dynamics recover the nominal LIP dynamic, so the H-LIP step-planner from [1] can be carried over in shifted coordinates:

Desired pre-impact states as:

$$x^{-*} = x_{eq} + \frac{u^*}{2 + \sigma T_d};\ v^{-*} = \sigma \frac{u^*}{2 + \sigma T_d}$$

, where $\sigma = \alpha \coth\left(\frac{\alpha T s}{2}\right)$

Nominal PD stepping controller using current pre-impact error to update next step length:

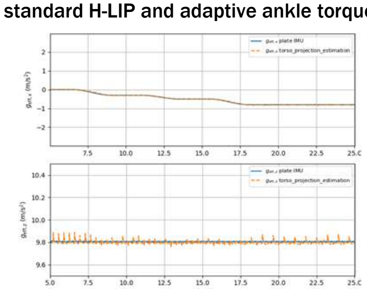
$$u_k = u^* + K \begin{bmatrix} x_k^- - x^{-*} \\ v_k^- - v^{-*} \end{bmatrix}$$

Implementation note: For onboard use, g_{eff} is estimated from torso IMU acceleration and stance-foot kinematics.

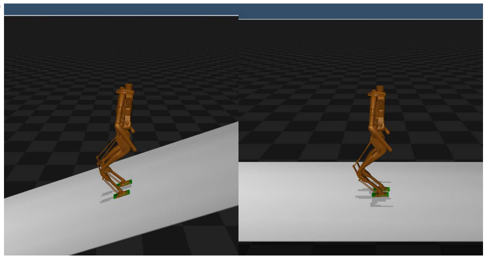
Simulation Setup

We evaluate the controller in MuJoCo on a 36-DOF Digit humanoid walking on a controllable platform. The method is benchmarked against standard H-LIP and adaptive ankle torque control.

Disturbance profiles with step changes use 2.5 s transition ramps to avoid physically unrealistic discontinuities.



Signal tracking of proposed Onboard Sensing



Simulation samples of Biased H-LIP on a 15° slope and a 1 m/s² accelerating platform, comparable to bus-level acceleration.

Simulation results

Case	DRS profile	Standard H-LIP (m/s)	Adaptive Ankle Torque (m/s)	Biased H-LIP (m/s)
1	$0\ \mathrm{m/s^2}$	0.024	0.024	0.024
2	$0\ (4.5\mathrm{s}) \to -0.2\ (4.5\mathrm{s}) \to -0.4\ (9\mathrm{s})\ \mathrm{m/s^2}$	0.105	0.030	0.027
3	$0\ (4.5\mathrm{s}) \to 0.4\ (4.5\mathrm{s}) \to 0.8\ (9\mathrm{s})\ \mathrm{m/s^2}$	0.236	Failed	0.034
4	$0\ (4.5\mathrm{s}) \to 0.5\ (9\mathrm{s}) \to 0\ (4.5\mathrm{s})\ \mathrm{m/s^2}$	0.122	Failed	0.029
5	$x(t) = 0.25\left(1 - \cos\left(\frac{2\pi(t-2.0)}{5}\right)\right)\ \mathrm{m},\ t \text{ in s}$	0.107	0.046	0.053
6	$0^\circ \to 3^\circ \to 6^\circ$	0.274	0.020	0.027
7	$0^\circ \to -9^\circ \to -17^\circ$	Failed	0.078	0.047
8	$0^\circ\ (4.5\mathrm{s}) \to -8^\circ\ (9\mathrm{s}) \to 0^\circ\ (4.5\mathrm{s})$	0.338	0.048	0.022
9	$p(t) = 5^\circ\ (1 - \cos(0.4\pi t)),\ t \text{ in s}$	Failed	0.030	0.033

RMS Velocity Tracking Error Across Controllers.

Key Findings

Overall Evaluation

- The biased H-LIP controller achieved the most consistent performance across the DRS test cases.
- The method performed especially well under larger or sustained acceleration and slope disturbances.
- Adaptive ankle torque performed better in some small, continuously changing disturbance cases, which is expected because biased H-LIP corrects motion through steps, so recovery may require one or more steps.

Takeaway:

Biased H-LIP improves overall robustness across disturbance types. Although step-based recovery can be slower for small transient disturbances, our formulation provides a model-based interpretation of how slope and acceleration shift CoM behavior, which can guide future robust control or RL-based stabilization methods.

[1] X. Xiong and A. D. Ames, "3-D underactuated bipedal walking via H-LIP based gait synthesis and stepping stabilization," IEEE Trans. Robot., vol. 38, no. 4, 2022.

[2] Z. Li, C. Zhou, N. G. Tsagarakis, and D. G. Caldwell, "Compliance control for stabilizing the humanoid on the changing slope based on terrain inclination estimation," Auton. Robots, vol. 40, no. 6, 2016.

[3] J. Stewart, I.-C. Chang, Y. Gu, and P. A. Ioannou, "Adaptive ankle torque control for bipedal humanoid walking on surfaces with unknown horizontal and vertical motion," arXiv:2410.11799, 2024.